

POWER SYSTEMS – II LAB

Manual Calculations – Experiments 1 to 15

10-mark format: given data, formula, substitution, calculation and final result. Based on the uploaded PS-II Lab Manual.

Note: Where the manual provides only a simulation setup and no numerical output, the calculation uses only the numerical data stated in the manual and does not invent simulation values.

EXPERIMENT 1

Daily Load Curve and Load Duration Curve

Given

24-hour load data = [30, 35, 32, 40, 50, 65, 80, 90, 85, 75, 70, 65, 60, 65, 70, 85, 100, 110, 95, 80, 70, 55, 45, 35] kW. Connected load = 150 kW.

1. Maximum demand

Maximum Demand = $\max(\text{load}) = 110$ kW.

2. Average load

Average Load = $\Sigma P / 24 = 1587 / 24 = 66.125 \approx 66.12$ kW.

3. Load factor

LF = Average Load / Maximum Demand = $66.125 / 110 = 0.6011 = 60.11\%$.

4. Demand factor

DF = Maximum Demand / Connected Load = $110 / 150 = 0.7333 = 73.33\%$.

FINAL RESULT

Maximum demand = 110 kW; average load = 66.12 kW; load factor = 60.11%; demand factor = 73.33%.

EXPERIMENT 2

Linear Regression Load Forecasting

Given

Years X = 2018–2024. Loads Y = 120, 128, 135, 145, 150, 162, 170 MW.

1. Formula

$$\text{Load} = mX + c.$$

$$m = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / [n\Sigma X^2 - (\Sigma X)^2]$$

$$c = [\Sigma Y - m\Sigma X] / n.$$

2. Sums

$$n=7; \Sigma X=14147; \Sigma Y=1010; \Sigma XY=2041443; \Sigma X^2=28591115.$$

3. Slope

$$m = [7(2041443) - 14147(1010)] / [7(28591115) - 14147^2] = 8.3214 \text{ MW/year}.$$

4. Intercept

$$c = [1010 - 8.3214(14147)] / 7 = -16673.3214.$$

5. Equation

$$\text{Load} = 8.3214(\text{Year}) - 16673.3214.$$

6. Predictions

$$2025 = 177.57 \text{ MW}; 2026 = 185.89 \text{ MW}; 2027 = 194.21 \text{ MW}; 2028 = 202.54 \text{ MW}; 2030 = 219.18 \text{ MW}.$$

FINAL RESULT

Regression equation = $8.3214(\text{Year}) - 16673.3214$. Predicted 2030 load = 219.18 MW.

EXPERIMENT 3

Single-Area Load Frequency Control

Given

$R=2.4$; $D=0.8$; $H=5$ s; $f_0=50$ Hz; $T_g=0.08$ s; $T_t=0.3$ s; $PL=0.01, 0.02, 0.05$ pu.

1. Time constant

$$T_p = 2H/(Df_0) = 2(5)/(0.8 \times 50) = 0.25 \text{ s}.$$

2. Gain

$$K_p = 1/D = 1/0.8 = 1.25.$$

3. Transfer functions

$$G_g=1/(0.08s+1); G_t=1/(0.3s+1); G_p=1.25/(0.25s+1).$$

4. Open loop

$$G = 1.25/[(0.08s+1)(0.3s+1)(0.25s+1)].$$

5. Closed-loop DC gain

$$\text{Sys}(0)=1.25/[1+1.25/2.4] = 0.821918.$$

6. Steady-state deviations

$$\Delta f_{ss} = -PL \times \text{Sys}(0): \text{ for } 0.01 \text{ pu} \rightarrow -0.008219 \text{ Hz}; 0.02 \text{ pu} \rightarrow -0.016438 \text{ Hz}; 0.05 \text{ pu} \rightarrow -0.041096 \text{ Hz}.$$

FINAL RESULT

The model gives transient frequency variation after the disturbance and the above settled frequency deviations for the implemented feedback model.

EXPERIMENT 4

Speed Governing System and Droop

Given

$$R=0.05 \text{ (5\% droop)}; T_g=0.2 \text{ s.}$$

1. Governor model

$$G_{gov}(s)=1/(0.2s+1).$$

2. Droop formula

$$\Delta P = -(1/R)\Delta f = -20\Delta f.$$

3. Example: $\Delta f=-0.05$ pu

$$\Delta P = -20(-0.05) = 1.0 \text{ pu.}$$

4. Example: $\Delta f=-0.025$ pu

$$\Delta P = -20(-0.025) = 0.50 \text{ pu.}$$

5. Example: $\Delta f=-0.01$ pu

$$\Delta P = -20(-0.01) = 0.20 \text{ pu.}$$

FINAL RESULT

Droop relation is linear. At $R=0.05$, a -0.05 pu frequency deviation corresponds to $+1.0$ pu load change.

EXPERIMENT 5

Automatic Generation Control – Single Area

Given

$R=2.4$; $D=0.8$; $H=5$ s; $f_0=50$ Hz; $T_g=0.08$ s; $T_t=0.3$ s; $PL=0.02$ pu.

1. Generator-load parameters

$T_p=2H/(Df_0)=0.25$ s; $K_p=1/D=1.25$.

2. Models

$G_g=1/(0.08s+1)$; $G_t=1/(0.3s+1)$; $G_p=1.25/(0.25s+1)$.

3. DC gain

$Sys(0)=1.25/[1+1.25/2.4]=0.821918$.

4. Steady-state frequency deviation

$\Delta f_{ss}=-0.02 \times 0.821918=-0.016438$ Hz.

FINAL RESULT

For the feedback model implemented in the manual, the 0.02 pu step gives a calculated steady-state frequency deviation of approximately -0.01644 Hz.

EXPERIMENT 6

Two-Area Load Frequency Control

Given

$R_1=R_2=2.4$; $D_1=D_2=0.8$; $T_{g1}=T_{g2}=0.08$ s; $T_{t1}=T_{t2}=0.3$ s; $K_{p1}=K_{p2}=120$; $T_{p1}=T_{p2}=20$ s; $T_{12}=0.545$; $P_L=0.01$ pu.

1. Area model

$G_1=120/[(0.08s+1)(0.3s+1)(20s+1)]$. Area 2 has the same model.

2. Closed-loop DC gain

$\text{Sys}(0)=120/[1+120/2.4]=2.307692$.

3. Area 1

$\Delta f_1, ss=-0.01 \times 2.307692=-0.023077$ Hz.

4. Area 2

The MATLAB program applies $0.5P_L$: $\Delta f_2, ss=-0.005 \times 2.307692=-0.011538$ Hz.

5. Tie line

$\Delta P_{tie}=T_{12}(\Delta f_1-\Delta f_2)=0.545(-0.023077+0.011538)\approx-0.00629$ pu.

FINAL RESULT

Area-1 and Area-2 frequency deviations and tie-line deviation are obtained from the simplified transfer functions used in the manual.

EXPERIMENT 7

Automatic Voltage Regulator (AVR)

Given

$K_a=10$, $T_a=0.1$ s; $K_e=1$, $T_e=0.4$ s; $K_g=1$, $T_g=1$ s; $K_s=1$, $T_s=0.05$ s; $V_{ref}=1$ pu.

1. Component models

$G_a=10/(1+0.1s)$; $G_e=1/(1+0.4s)$; $G_g=1/(1+s)$; $G_s=1/(1+0.05s)$.

2. Forward path

$G_{forward}=G_a G_e G_g$.

3. Closed loop

$T(s)=G_{forward}/[1+G_{forward}G_s]$.

4. Simple DC-gain calculation

At $s=0$: $G_{forward}(0)=10$ and $G_s(0)=1$. Therefore $T(0)=10/(1+10)=0.9091$ pu.

5. Manual simulation output

The manual reports terminal voltage = 0.8023 pu and steady-state error = 19.77%.

FINAL RESULT

Use the MATLAB simulation output for the lab record: $V_{t,ss} = 0.8023$ pu and steady-state error = 19.77%. The simple DC-gain value is shown separately.

EXPERIMENT 8

Shunt Capacitor – Power Factor Improvement

Given

$V_{LL}=400\text{ V}$; $f=50\text{ Hz}$; $P=10\text{ kW}$; $Q_L=8\text{ kVAR}$; $L=10\text{ mH}$; target $PF=0.95$; selected capacitor= 5 kVAR .

1. Line reactance

$$X_L=2\pi fL=2\pi(50)(0.01)=3.142\ \Omega.$$

2. Initial apparent power

$$S_1=\sqrt{(10^2+8^2)}=12.806\text{ kVA}.$$

3. Initial PF

$$PF_1=10/12.806=0.781\text{ lagging}.$$

4. Initial current

$$I_1=12806/(\sqrt{3}\times 400)=18.48\text{ A}.$$

5. Target reactive power

$$Q_{\text{target}}=10 \tan(\cos^{-1}0.95)=3.287\text{ kVAR}.$$

6. Capacitor required

$$Q_C=8-3.287=4.71\text{ kVAR}.$$

7. With 5 kVAR capacitor

$$Q_{\text{new}}=8-5=3\text{ kVAR}; S_2=\sqrt{(10^2+3^2)}=10.440\text{ kVA}; PF_2=10/10.440=0.958; I_2=10440/(\sqrt{3}\times 400)=15.07\text{ A}.$$

8. Current reduction

$$[(18.48-15.07)/18.48]\times 100=18.47\%.$$

FINAL RESULT

Initial PF = 0.781; final PF = 0.958; current reduction = 18.47%; required capacitor $\approx 4.71\text{ kVAR}$.

EXPERIMENT 9

STATCOM / SVC Voltage Profile Study

Given

500 kV transmission system; STATCOM ± 100 MVAR; $V_{ref}=1.0$ pu; source disturbance=0.95–1.05 pu; simulation time=1.5 s.

1. STATCOM rating in pu

On a 100 MVA base: $Q_{max} = \pm 100/100 = \pm 1.0$ pu.

2. Voltage disturbance

$V_{source,min}=0.95$ pu; $V_{source,max}=1.05$ pu.

3. Voltage error

$e(t)=V_{ref}-V_{bus}(t)$. If $V_{bus}<1.0$ pu, capacitive VAR injection is required; if $V_{bus}>1.0$ pu, reactive absorption is required.

4. Operating relation

$V_{STATCOM}>V_{line} \rightarrow$ capacitive injection. $V_{STATCOM}<V_{line} \rightarrow$ inductive absorption.

FINAL RESULT

The manual is simulation-based and does not provide a numerical final bus-voltage/Q value. No numerical simulation result is invented here.

EXPERIMENT 10

DC Power System State Estimation using WLS

Given

$z=[0.51, 0.18, 0.315]$ pu; $H=[[5,0],[0,2.5],[-4,4]]$; $\sigma=0.01$ pu.

1. Covariance and weight

$R=\text{diag}(0.0001,0.0001,0.0001)$; $W=R^{-1}=10000I$.

2. WLS equation

$x=(H^TWH)^{-1}H^TWz$.

3. Estimated states

$\theta_1=0$ rad; $\theta_2=0.085429$ rad; $\theta_3=0.138286$ rad.

4. Estimated measurements

$z^*=Hx$: $P_{12}=0.427143$ pu; $P_{13}=0.345714$ pu; $P_{23}=0.211429$ pu.

5. Residuals

$r=z-z^*$: $r_1=0.082857$ pu; $r_2=-0.165714$ pu; $r_3=0.103571$ pu.

6. Maximum residual

$\max|r|=0.165714$ pu.

FINAL RESULT

WLS state estimation gives the above bus angles, estimated measurements and maximum residual for the manual data.

EXPERIMENT 11

SCADA and Remote Terminal Unit (RTU)

Given

Manual problem: 132 kV feeder, normal load=80 MW; communication delay=2 s. MATLAB demonstration uses 11 kV and 50 MW.

1. Normal condition

Breaker CLOSED; voltage and load are sent as telemetry to the Master Station.

2. Trip condition in the MATLAB model

Command=0 → Breaker OPEN; Load=0 MW; Bus Voltage=0 kV.

3. Time-stamp difference

Δt = communication delay = 2 s.

4. Re-close condition

Command=1 in the MATLAB demonstration → Breaker CLOSED; Load=50 MW; Bus Voltage=11 kV.

FINAL RESULT

For the stated delay, the HMI visualization is 2 s after the actual event. The manual's 132 kV/80 MW problem does not specify a post-trip numerical voltage.

EXPERIMENT 12

Bad Data Detection using Chi-Square Test

Given problem data

$m=5$ measurements; $n=3$ states; $J(x)=12.5$; $\chi^2_{\text{critical}}=5.99$ for $k=2$ at 95% confidence.

1. Degree of freedom

$$k=m-n=5-3=2.$$

2. Test condition

$$J(x)=12.5 > 5.99=\chi^2_{\text{critical}}.$$

3. Decision

Because J exceeds the stated threshold, bad data is detected.

4. Corrective action

Identify the measurement with the largest normalized residual, remove/correct it, and repeat state estimation.

5. MATLAB data

$z_{\text{true}}=[0.5, 0.2, 0.3]^T$; $\sigma=0.01$; injected error= $[0.001, 0.6, -0.002]^T$; therefore
 $z_{\text{noisy}}=[0.501, 0.800, 0.298]^T$.

6. Objective function

$$J(x)=r^T W r = \sum [(z_i - h_i(x))^2 / \sigma_i^2].$$

FINAL RESULT

The manual's numerical test gives $k=2$ and $12.5 > 5.99$, so bad data exists. The later printed normalized-residual identification in the manual is internally inconsistent, so it is not reproduced as a mathematical conclusion.

EXPERIMENT 13

Economic Dispatch – Two Thermal Units

Given

PD=300 MW. U1: 50–250 MW, $a_1=500$, $b_1=5$, $c_1=0.01$. U2: 50–200 MW, $a_2=400$, $b_2=6$, $c_2=0.015$.

1. Cost functions

$$F_1=500+5P_1+0.01P_1^2; F_2=400+6P_2+0.015P_2^2.$$

2. Incremental cost equality

$$5+0.02P_1 = 6+0.03P_2.$$

3. Power balance

$$P_1+P_2=300 \rightarrow P_2=300-P_1.$$

4. Solve

$$5+0.02P_1=6+0.03(300-P_1) \rightarrow 0.05P_1=10 \rightarrow P_1=200 \text{ MW}.$$

5. Unit 2 output

$$P_2=300-200=100 \text{ MW}.$$

6. Incremental costs

$$IC_1=5+0.02(200)=9; IC_2=6+0.03(100)=9 \text{ cost/MWh}.$$

7. Costs

$$F_1=500+1000+400=1900 \text{ cost/h}.$$

$$F_2=400+600+150=1150 \text{ cost/h}.$$

8. Total

$$\text{Total cost}=1900+1150=3050 \text{ cost/h}.$$

9. Limits

$$50 \leq 200 \leq 250; 50 \leq 100 \leq 200; 200+100=300 \text{ MW}.$$

FINAL RESULT

$P_1=200$ MW; $P_2=100$ MW; equal incremental cost=9 cost/MWh; total generation cost=3050 cost/h.

EXPERIMENT 14

Unit Commitment – Simplified Two-Unit System

Given

U1: $P_{\min}=100$, $P_{\max}=400$ MW; $C_1=0.01P_1^2+20P_1+100$; startup=500. U2: $P_{\min}=50$, $P_{\max}=300$ MW; $C_2=0.02P_2^2+15P_2+150$; startup=300. Loads=[150, 450, 600, 200] MW.

1. Hour 1: 150 MW

U2 alone: running cost= $0.02(150^2)+15(150)+150=2850$; startup=300; total=3150. U1 alone and both units are also capacity-feasible, but the supplied MATLAB output selects U2 alone.

2. Hour 2: 450 MW

U1 alone: $450>400$, infeasible. U2 alone: $450>300$, infeasible. Both units: simplified ED gives $P_1\approx 133.33$ and $P_2\approx 316.67$ MW, so U2 exceeds 300 MW and is rejected.

3. Hour 3: 600 MW

U1 alone and U2 alone are infeasible. Both units give simplified ED $P_1\approx 233.33$ and $P_2\approx 366.67$ MW; U2 exceeds 300 MW, so rejected.

4. Hour 4: 200 MW

U2 alone: running cost= $0.02(200^2)+15(200)+150=3950$; startup=300; total=4250.

FINAL RESULT

The manual output shows Inf for Hours 2 and 3 because the simplified program rejects the combinations after generator-limit checks. This follows the supplied manual rather than replacing its algorithm.

EXPERIMENT 15

Demand Side Management (DSM) – Load Shifting

Manual problem data

Peak=1200 MW; valley=400 MW; shift=150 MW; peak cost=7429.20/MWh; off-peak cost=3714.60/MWh.

1. New peak

New peak=1200-150=1050 MW.

2. New valley

New valley=400+150=550 MW.

3. Average load

Shifting does not change daily energy, so average load remains P_{avg} . New LF= $P_{avg}/1050$.

4. Original LF

Original LF= $P_{avg}/1200$.

5. Relative improvement

New LF / Original LF = $1200/1050 = 1.142857$. Thus the new LF is 14.29% higher relative to the original LF.

6. Daily saving for a 1-hour shift

Cost difference=7429.20-3714.60=3714.60/MWh. Shifted energy=150 MW×1 h=150 MWh.

Saving=150×3714.60=557,190 per day.

7. MATLAB profile note

The code uses a different example: original peak=1000 MW, shift=100 MW, new peak=900 MW; original average=661.67 MW and new average=661.67 MW; LF changes from about 0.6617 to 0.6965.

FINAL RESULT

For the stated problem, new peak=1050 MW and new valley=550 MW; a 1-hour 150-MW shift gives 557,190 per day in calculated cost difference. Exact new load factor requires the original average load, which the problem statement does not provide.